

Combining low-loss EELS experiments with machine learning-based algorithms to automate the phases separation imaging in industrial duplex stainless steels

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ARTICLE INFO

Keywords:

Clustering

Low-loss EELS

Plasmon

Phase identification

Duplex stainless steels

ABSTRACT

At present, transmission electron microscopy is regarded as the main option when dealing with phase characterization for materials at a nanometric scale. The development and improvement of complementary techniques such as energy-dispersive X-ray spectroscopy (EDS), electron energy loss spectroscopy (EELS), imaging detectors and associated computational methods provide a huge variety of choices to determine the composition and crystal structure at any region of a specimen. Despite all these advancements, phase identification for some specific materials having phases with similar structure or similar composition might still be a difficult procedure to perform. To overcome this difficulty, a new method is proposed which combines the acquisition of low-loss EELS spectra with a subsequent analysis driven by machine learning based algorithms (clustering). In the present research, this new approach is applied to a set of industrial duplex stainless steels having different compositions and it is contrasted to other characterization alternatives. As opposed to the other options, this method proved to be effective, reliable and time-saving.

1. Introduction

Duplex stainless steels (DSS) constitute a family of steels made of chromium-nickel-molybdenum-iron bi-phased alloys in which α -ferrite and γ -austenite fractions are present in relatively large separate volumes. The first commercial DSS grades were produced around 1930 by Avesta Steelworks and were mainly used in the pulp and paper industry [1]. Nowadays with the development of lean and super-duplex grades, DSS cover a wide range of applications including pipes and storage tanks for the chemical and petrochemical industries and heat exchangers and umbilicals for seawater environments [1]. In addition, they are also used in construction works and vehicles such as bridges, tankers as well as architectural buildings among which we can cite *La Sagrada Familia* in

Barcelona [1–3]. Due to their bi-phased microstructure, DSS possess higher mechanical strength and better corrosion resistance than standard austenitic stainless steels and are cost savings due to their lower nickel content. However, during aging a large variety of secondary phases, including carbides, nitrides, χ , σ and Laves phases, are known to precipitate in DSS and degrade their properties². The σ phase, which is an intermetallic compound richer in Cr and Mo than the surrounding matrix, is considered to be the most detrimental of all phases due to its potential to form in significant amount [4]. The σ phase is a hard and brittle phase which decreases the toughness and ductility of DSS. In addition, the formation of σ precipitates leads to a depletion of Cr and Mo elements from the surrounding environment leading thus to a reduction in corrosion resistance of the DSS [4–6]. In most of DSS, the

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formation of the σ phase is reported in the temperature range 600–1000 °C with holding times which can be as low as few minutes [2,4,5]. It occurs mostly at the austenite/ferrite interface, and to a lesser extent, at the ferrite/ferrite interface [2,7]. At the microscopic level, electron backscatter diffraction in a scanning electron microscope (SEM-EBSD) and transmission kikuchi diffraction (SEM-TKD) are the favored techniques for examining the phases in presence in such materials. However, their effectiveness is constrained by the spatial resolution of the SEM. One of the primary challenges arises from the need for a technique capable of concurrently imaging interfaces spanning several tens of microns, such as austenite/ferrite interfaces, while also addressing more localized interfaces associated with secondary phase precipitates, which may be restricted to 100 nm and below. Characterizing and mapping the interphases and phases in presence of as-cast and aged DSS at the nanoscale is thus of high-importance. In addition, in an industrial context like in the present work, data acquisition and analysis should be performed in a timely manner. Among all the available methods which can be used for phase identification, low-loss EELS experiments in a transmission electron microscope can be acquired from μm to nm scales and allow to obtain large dataset within short acquisition time. However, interpretation and analysis of such data is not straightforward. Low-loss EELS spectra contain many excitation processes including exciton, interband transition, surface and volume plasmon [8]. Volume plasmon can be used for fingerprinting approaches but it requires the use of a catalogue of reference spectra and laborious pixel per pixel fitting procedures⁸. Precession electron diffraction (PED) in a transmission electron microscope (TEM) offers the capability to obtain structural details at the nanometer scale. However, unlike techniques such as electron energy loss spectroscopy (EELS), PED is not as widely available across microscopes and may pose greater challenges during setup compared to low-loss EELS experiments. Furthermore, one day of experiment with an advanced (S)TEM microscope in which we combine several imaging and spectroscopic experiments can yield a high amount of data which, depending on their complexity, can take several days or even weeks to interpret. With the on-going development of in-situ microscopy, this trend will increase and it is estimated that in-situ (S)TEM generates an amount of data higher by 3–4 orders of magnitude than ex-situ [9]. It is thus necessary to create automatic tools of analysis that can extract statistical and physically meaningful information. For this purpose, the EELS community already transitioned to the era of big data [10–12]. Maybe the most common example is the use of principal component analysis (PCA) for denoising EELS dataset. PCA, independent component analysis (ICA) and non-negative matrix factorization (NMF) can also be used to cluster the data, i.e., to group different sets of objects by similarity. However, when applied to EELS and even EDS data, PCA, ICA and NMF have strong difficulties to output components with physical meaning. On the other hand, clustering approaches have been more successful to extract information with physical meaning when applied to core-loss EELS dataset [13–15]. In the present work, we explored the possibility offered by machine learning based techniques, particularly k-means clustering, on low-loss EELS experiments in order to automatically separate the phases in presence of industrial as-cast and aged industrial DSS. It allowed us, not only to map α and γ phases, but also intermetallic phases such as the σ phase in quick fashion manner.

The reliability and efficiency of this new approach is compared to more traditional methods for phase recognition like Z-contrast in STEM-HAADF, EDS compositional mappings and diffraction patterns (DPs) identification. The advantages and disadvantages in each case are summarized. The validity of clustering method is also analyzed against a pixel per pixel fitting based on Drude's model that allows to map the plasmon energy and obtain the plasmon width parameters. Additionally, the interpretation of low-loss EELS spectra is supported by calculations based on Density-Functional Theory (DFT).

2. Experimental and methods

2.1. Alloy synthesis

As-cast duplex stainless steels alloys with various Ni content (2205, 2205 M1, 2304 and 2001) were fabricated at the ACERINOX EUROPA SAU plant in Algeciras. Raw materials were melted in a Pfeiffer-Balzars VSG030 induction furnace and the subsequent casting was performed in controlled atmosphere to obtain 40 kg ingots. The composition of the samples was provided by the fabricant and is displayed in the Supplementary Information (Table S1) as well as the amount of ferrite and austenite phases obtained for each ingot (Table S2). These DSS alloys were tailored with the goal of obtaining approximately the same amount of ferrite and austenite phases, but with different Ni content. Four ingots are obtained from: 2205 and 2205 M1 alloys ($C_{\text{Ni}} = 5.4 \text{ wt\%}$), 2304 alloy ($C_{\text{Ni}} = 4 \text{ wt\%}$) and 2001 alloy ($C_{\text{Ni}} = 1.85 \text{ wt\%}$).

Additionally, “ductility tests” or “hot deformation tests” were conducted at 1090 °C in all of these alloys. Hot deformation tests can provide important information regarding the material behavior under extreme temperature conditions, thus supporting proper material selection and manufacturing process choices for a given application. In this kind of tests, the sample is subjected simultaneously to a tensile stress and a high temperature environment, and the degree of deformation is measured. This procedure usually leads to the formation of precipitates from secondary phases in steels, like those belonging sigma to phase.

2.2. TEM sample preparation and observation

TEM samples were prepared by electro-chemical polishing using a double jet Tenupol-5 device and A2 electrolyte, both from Struers. The working temperature for this procedure was around +5 °C. Previous to this step, discs with a diameter of 3 mm and a thickness of around 100 μm were obtained by mechanical grinding and a disc punch. Later on, the TEM foils were studied in a FEI Titan Cubed Themis 60–300 microscope operated at 200 kV and equipped with a double Cs aberration-corrector, a monochromator, an X-FEG gun, an Ultra High Resolution Energy Filter (Gatan Quantum ERS) which allows working in dual-EELS mode and a Super X EDS detector. It should be noted that the EELS experiments were performed without exciting the monochromator. Imaging was performed by using high-angle annular dark field (HAADF) and ADF detectors. The convergence angle is 16 nm for the all the experiments. Most of the STEM-HAADF images were acquired by using values corresponding to 50 and 200 mrad for the inner and outer angles, respectively. Absorption correction for EDS quantification was performed by taking into account the thickness of the probed area. For this purpose, low-loss EELS measurements were used to determine the t/λ ratio (t being the thickness of the analyzed crystal and λ the inelastic mean free path) and the modified Iakubovskii formula was used to determine λ [16,17]. Selected area electron diffraction (SAED) patterns were obtained and compared to simulated diffraction patterns. The simulations were carried out by means of jems4 software [18]. Automated indexation of the SAED patterns was performed by the software to determine corresponding phase and crystal orientation. EELS data were acquired with a dispersion of 0.1 eV/pixel and an energy resolution of 0.8 eV. Dataset of about $18 \mu\text{m} \times 18 \mu\text{m}$ (corresponding to 100 pixels $\times$ 100 pixels) were acquired with a dwell time of 0.05 s. Next, the single scattering spectra were obtained by using the Fourier ratio method [17].

2.3. Clustering of EELS spectra

After multiple scattering removal, EELS datasets were clustered by using a python's machine learning based clustering algorithm called K-means from Scikit-Learn library implemented in the hyperspy python package. The K-means algorithm attempts to classify objects into a pre-

determined number of clusters by finding optimal cluster centers (also called centroids) for each cluster. Each object is then classified as belonging to the cluster represented by the closest centroid [19]. The only parameter needed, previous to clustering, was the number of clusters which was obtained by evaluating the silhouette metric [20]. The silhouette metric helps in assessing the compactness and separation of clusters and higher silhouette scores indicate better clustering quality.

2.4. Pixel per pixel fitting

Results from clustering were compared with a classical numerical method that fits the EELS datasets pixel per pixel by using a Drude model. For this purpose, the $(\pi + \sigma)$ volume plasmon was modeled over the top 80% of the plasmon peak intensity in order to extract the plasmon energy and width [21]. More details are given in Section III.a from Supplementary Information.

2.5. DFT calculations

In order to interpret the results, EELS experimental spectra were also compared with calculated spectra based on the density-functional theory (DFT) method. Spin-polarized density functional theory calculations were performed using the Vienna Ab Initio Simulation Package (VASP) [22,23]. The projector-augmented wave (PAW) method [24,25] was used to represent core and valence electrons while electron exchange and correlation were modeled using the Perdew-Burke-Ernzerhof functional [26]. Pure BCC and FCC iron were modeled using primitive cells corresponding to (conventional) lattice parameters of 2.863 Å and 3.655 Å, respectively [27]. For both phases, the kinetic-energy cutoff was set to 550 eV, Brillouin-zone integrations were performed using the second-order Methfessel-Paxton method [28] with a smearing width of 0.2 eV, and a $51 \times 51 \times 51$ Monkhorst-Pack k -point mesh was used to sample the Brillouin zone [29]. A complex shift of 0.1 eV was used in linear response calculations. From the frequency-dependent dielectric function, $\epsilon(\omega)$, in the long-wavelength limit ($q \rightarrow 0$), the EELS intensity

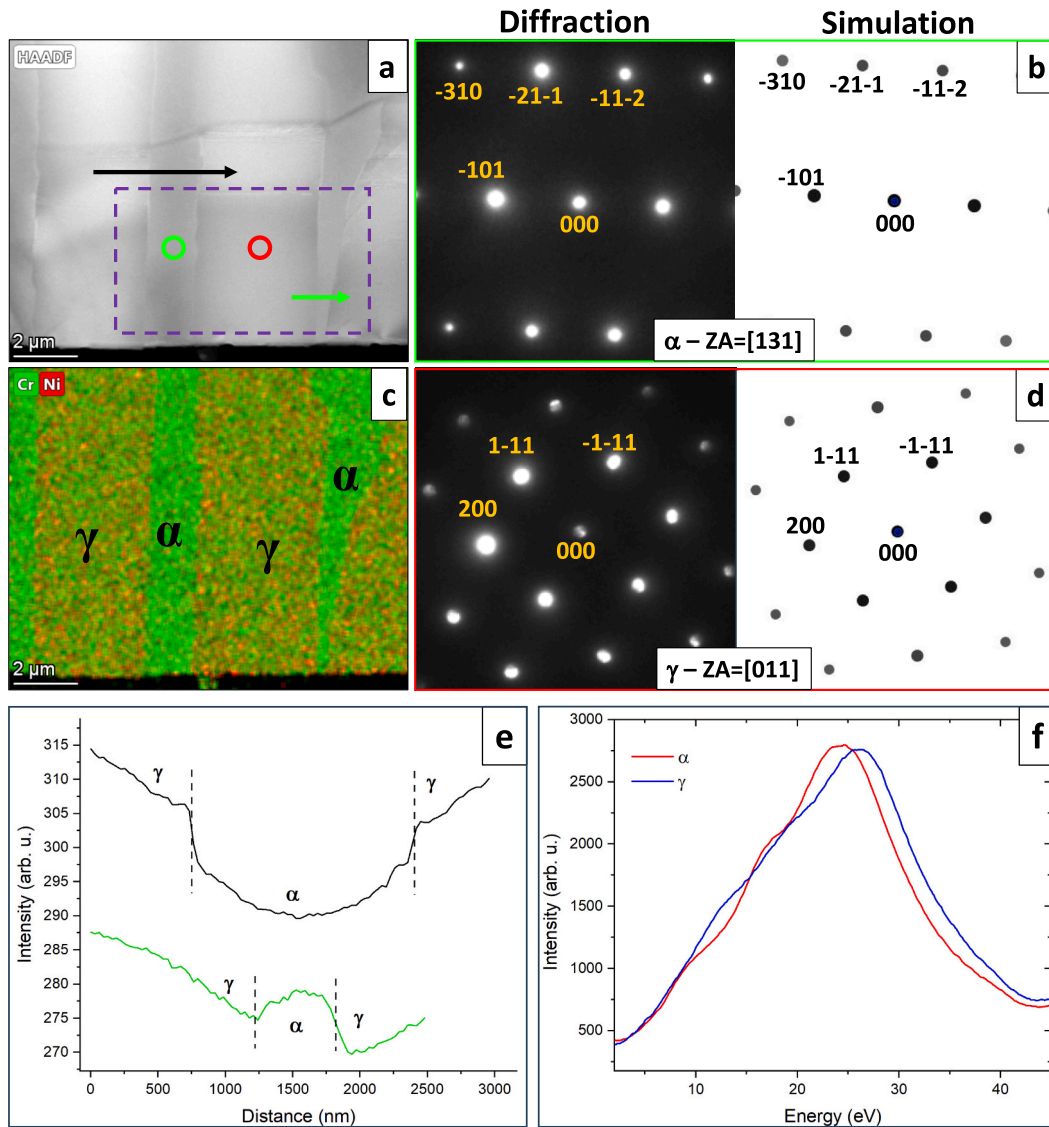


Fig. 1. 2205 steel, as-cast: a) STEM-HAADF micrograph. Red and green circles highlight the area where the SAED patterns were obtained. b) SAED pattern performed in green circle region (left) and its matching simulated diffraction pattern (right) which corresponds to α ferrite ($Im-3m$) observed along the [131] direction. c) Cr and Ni EDS chemical maps. d) SAED pattern acquired in red circle region (left) which fits well with simulated DP from γ austenite ($Fm-3m$) in $ZA = [011]$ condition (right). Both, diffraction patterns and simulations have a size of $20 \text{ nm}^{-1} \times 20 \text{ nm}^{-1}$. e) Intensity profiles extracted in the places pointed by black and green arrows in (a). f) Low-loss EELS spectra acquired on the α and γ phases. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

was calculated as $\text{Im}[-1/\epsilon(\omega)]$ (up to an arbitrary prefactor).

3. Results and discussion

3.1. Traditional methods for phase Identification vs. low-loss EELS spectroscopy

A general exploration of 2205 in as-cast state was performed by means of STEM-HAADF imaging, and is shown in Fig. 1a. A lamellar microstructure is observed with alternating ferrite (α -phase) and austenite (γ -phase) regions. Diffraction patterns within two adjacent regions were obtained (Fig. 1b and Fig. 1d, left) and compared to simulated diffraction pattern (Fig. 1b and Fig. 1d, right) in order to determine nature of the phase in each lamellar column. For the SAED pattern shown in Fig. 1b, the area used to acquire the diffraction pattern is highlighted by a green circle in Fig. 1a. Similarly, for the SAED pattern shown in Fig. 1d, the area used to acquire the diffraction pattern is highlighted by a red circle in Fig. 1a. On the other hand, the EDS compositional determination and mapping was also obtained and the distribution of Cr and Ni in the duplex alloy is shown in Fig. 1c.

The indexation obtained from comparing the SAED pattern to simulated diffraction patterns confirms that the Ni-rich area (pointed by a red circle in Fig. 1a) corresponds to the γ austenite $Fm-3m$ structure [30] seen along the [011] zone axis, whereas the place showing the lowest Ni concentration (green circle region in Fig. 1a) corresponds to the α ferrite $Im-3m$ structure [31] seen along the [131] zone axis. It should be noted that in this case, it does not correspond to any preferential orientation relationship between the grains. Because of this, it's crucial to note that tilting the sample to attain a condition where, for example, all alpha regions align with a strong diffraction orientation (similar to a Zone Axis condition), while keeping gamma regions in a cinematic state, isn't feasible for this material.

Intensity line profiles obtained in two different places from 2205 specimen's HAADF image (Fig. 1a) are presented in Fig. 1e. The HAADF contrast is sensitive to atomic number (Z contrast) and to TEM foil local thickness, as well. In particular, an increase of the HAADF intensity as function of the Ni content could be expected. Thus, it is interesting in this case to evaluate if Z contrast can effectively discriminate between α and γ phases. With this aim, two intensity line profiles are acquired both completely traversing a ferritic region ($\gamma/\alpha/\gamma$ interfaces). The specific place in the STEM micrograph from where each profile is obtained is signaled by black and green arrows, respectively, within Fig. 1a. It can be observed that the black line intensity profile diminishes its intensity when crossing from the γ region to the α phase region. On the contrary, for the green line profile the opposite behavior takes place when traversing from γ to α regions, meaning that STEM-HAADF contrast exploration is not an appropriate method to discriminate these two phases. The reason for the unreliability of Z contrast in this type of alloy is due to the minor difference between the average Z number of α and γ phases ($Z_{\alpha}^{-} \sim 25.5$ and $Z_{\gamma}^{-} \sim 25.7$) as calculated from each phase composition given by EDS analyses, even while 2205 is the DSS having the highest Ni content of the series. In addition, the probed area shows variations of relative thickness which hinder the proper phase separation by using the analysis of the HAADF contrast.

As expected for alloy 2205 which is obtained from a DSS having a high Ni content ($C_{Ni} \sim 5.4$ wt%, Table S1) some regions are found to have a higher Ni content than others. In the regions identified by their SAED pattern to be γ the Ni concentration rise up to 6.1 at.%, while in the α regions Ni concentration is just around 3 at.%. Regarding the other elements of the alloy, more differences between γ and α can be noticed. For instance, γ presents a higher Fe concentration and a lower Cr content ($C_{Fe} \sim 66.2\%$, $C_{Cr} \sim 22.3\%$, at.%) than α ($C_{Fe} \sim 62.3\%$, $C_{Cr} \sim 27\%$, at.%) which is reflected in corresponding EDS maps (Fig. 1c).

The composition of the other three specimens (2304, 2001 and 2205 M1) were investigated in this same manner and the results from quantifying the concentration (in at. %) of the most relevant elements in each

phase are displayed in Table 1, together with the results obtained on 2205 specimen. It should be pointed out that some elements like S, Sn, Sb, P, Pb or Ti contributing <2 at.% in the whole composition are not detailed here.

Low-loss EELS spectra were acquired on the same area to determine the influence of the crystallographic structure and composition on the EELS response. The corresponding spectra of the α and γ phases are presented in Fig. 1f. The volume plasmon of the α and γ phases are clearly separated by about 2.0 eV. The maximum value of the plasmon (i.e., the absolute maximum) being at $E_p^{\alpha} = 24.20 \pm 0.07$ eV for α phase, and $E_p^{\gamma} = 26.20 \pm 0.14$ eV for γ phase. In addition, broad features at lower energy (i.e., relative local maxima) can be observed at about 8.2 and 16.8 eV for the α phase, and 12.7 and 19.3 eV for the γ phase, respectively.

Fig. 2 shows the similar STEM analyses performed on 2001 alloy which is the least alloyed DSS of the family, with a bulk Ni concentration of about 1.9 wt% (Table S1). Fig. 2a displays a HAADF micrograph while Fig. 2b portrays the corresponding Cr and Ni distribution. Most of the grains are elongated and have a width typically between 700 nm and 3 μm , and a length between 2 and 10 μm . STEM-HAADF imaging necessitates both an inner detector angle which is at least three times larger than the aperture angle and an angular width of about 50 mrad [32]. For this image, the inner and outer angles during the acquisition were 50 and 200 mrad, respectively, maintaining the expected ratio (3.13 in this instance). Despite this, diffraction contrast remains present in the STEM-HAADF micrograph and dislocation lines can be observed inside the grains. Once again, the Ni and Cr elemental maps present reversed relation between phases, showing higher Ni and lower Cr content for γ than for α .

From the image in Fig. 2a, an intensity profile is obtained within a single-phase region (α), in the place marked by a blue vertical arrow. This intensity profile is presented in Fig. 2c and it shows a pronounced decay in intensity across a length of around 500 nm but in absence of a phase change, proving once again the unreliability of this method to determine the limits of each phase domain or even the existence of different phases. On the contrary, each one of the present phases can still be clearly identified by its low-loss EELS spectrum (Fig. 2d) which preserve invariable shape and peak position with respect to those observed in the specimen from 2205 alloy (Fig. 1f).

For the sample obtained from alloy 2001, it is observed in Table 1 that Ni concentration (in at.%) increases from 0.85% in α to a value of 1.65% within the γ region, thus γ almost duplicates α 's Ni content. The Fe concentration remains almost unchanged (68.6% in α , 70.9% in γ), while the Cr content decreases from 24.3% in α to a value of 20.7% in γ . The composition of each phase yields similar average atomic numbers ($Z_{\alpha}^{-} \sim 25.4$ and $Z_{\gamma}^{-} \sim 25.5$) explaining then why inversion of the HAADF contrast can be observed within the same grain when diffraction contrast and/or thickness changes are taking over as the main phenomena behind the contrast variations (Fig. 2c).

To check any influence of the Ni content on the low-loss EELS response, EELS analyses have been performed again on this sample (Fig. 2d). The α and γ spectra acquired on the specimen from 2001 alloy are really similar in position and shape to the spectra acquired on the 2205 alloy, disregarding the differences in composition between these two alloys. In particular, the α and γ phases present maximum values of the volume plasmon at $E_p^{\alpha} = 24.10 \pm 0.25$ eV and $E_p^{\gamma} = 26.20 \pm 0.2$ eV, respectively. This suggests that low-loss EELS spectra obtained from these samples are better at discerning their crystallographic natures than their chemical compositions.

Similarly, HAADF-STEM, EDS compositional mappings and low-loss EELS experiments were carried out in 2304 and 2205 M1. Table 2 summarizes the different values of the maximum of the plasmon energy (E_p) and the nickel concentration (in at.%) of α and γ phases for the four different alloys of the as-cast series.

As observed in Table 1, the most remarkable characteristic regarding α and γ compositions in 2304 alloy is that it presents an intermediate Ni

Table 1

Elemental quantification (at. %) as determined by STEM-EDS analyses in each alloy.

Sample	2205		2304		2001		2205 M1	
Elements	γ -Aust.	α -Ferrite	γ -Aust.	α -Ferrite	γ -Aust.	α -Ferrite	γ -Aust.	α -Ferrite
Al	0.2	0.82	0.13	0.17	0.32	0.45	0.29	0.15
Si	0.72	0.84	0.66	1.07	0.64	0.86	0.78	0.87
V	0.24	0.28	0.2	0.24	0.25	0.28	0.23	0.35
Cr	22.32	26.98	23.3	28.25	20.7	24.33	22.45	28.42
Mn	2.03	1.73	2.17	1.81	5.27	4.28	2.45	2.06
Fe	66.21	62.34	69.22	66.01	70.93	68.64	66.24	63.09
Ni	6.1	3.03	3.53	1.68	1.65	0.85	6.02	2.84
Cu	0.22	0.25	0.36	0.27	0.01	0.02	0	0
Nb	0.09	0.03	0.13	0.15	0.02	0.03	0.07	0.08
Mo	1.12	1.89	0.15	0.19	0.12	0.2	1.12	2
Others	0.75	1.81	0.15	0.16	0.09	0.06	0.35	0.14

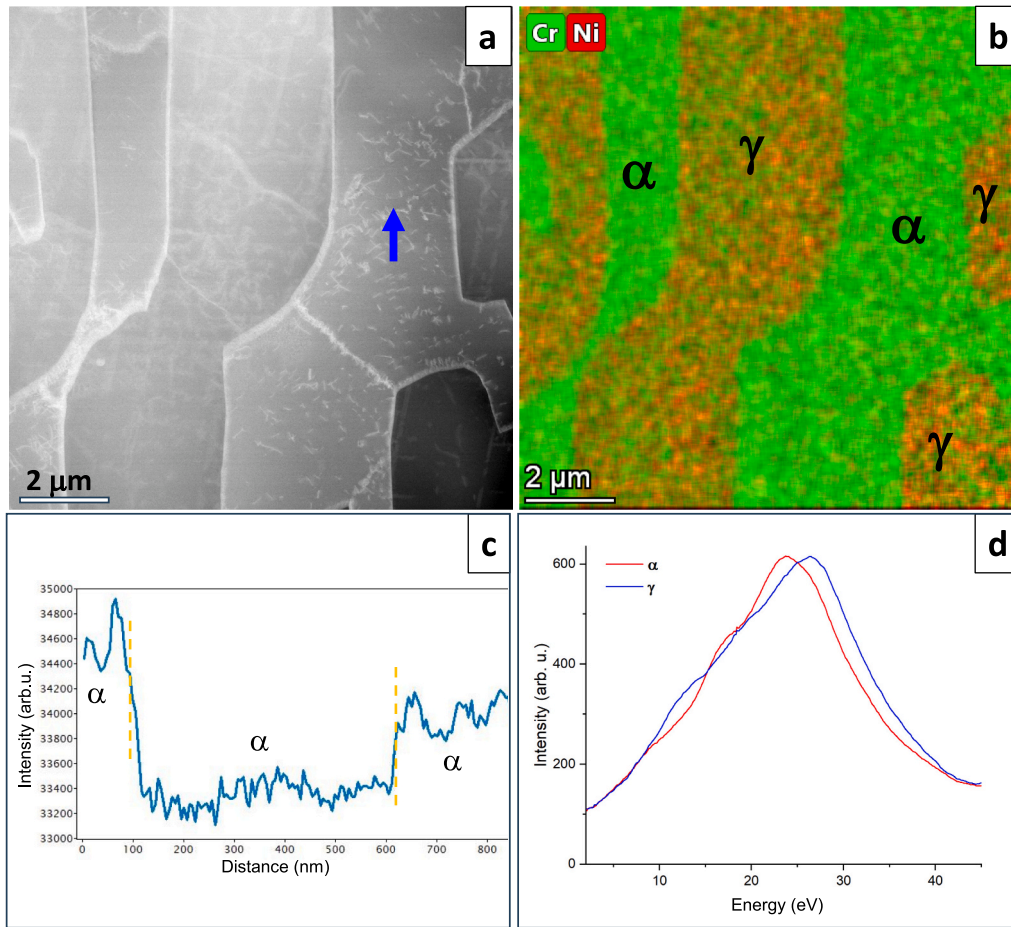


Fig. 2. **2001 alloy, as-cast:** a) HAADF-STEM image, b) EDS composition mapping of Cr and Ni distribution, c) Intensity profile extracted from the micrograph in (a), where the blue arrow is located. d) Low-loss EELS spectra acquired on the α and γ phases. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2

As-cast samples: Maximum energy position of the volume plasmon (E_p) for the α and γ phases of all the as-cast samples. The Ni content (C_{Ni}) in at.% as determined by EDS for each phase is also supplied. The errors correspond to standard errors determined by using the pixel per pixel fitting procedure which allows for sub-pixel accuracy.

Alloy	E_p^α (eV)	E_p^γ (eV)	C_{Ni}^α (at. %)	C_{Ni}^γ (at. %)
2205	24.20 ± 0.07	26.20 ± 0.14	3.0	6.1
2304	24.10 ± 0.31	26.10 ± 0.17	1.7	3.5
2001	24.10 ± 0.25	26.21 ± 0.20	0.9	1.7
2205 M1	24.20 ± 0.26	26.20 ± 0.28	2.8	6.0

content for both phases ($C_{Ni}^\alpha \sim 3.5\%$ and $C_{Ni}^\gamma \sim 1.7\%$, in at. %). Alloy 2205 M1 presents similar Ni concentrations than that observed 2205, but higher content of Mn. Nevertheless, as shown in Table 2, E_p in both phases remained unaffected by the compositional variations presented in each alloy. This could be attributed to minimal fluctuations in nickel content, which are unlikely to significantly alter the electron density surrounding the iron atoms.

3.2. DFT calculations

As the results in Table 2 suggest, the low-loss EELS response for DSS

seems to be most sensitive to the crystallographic structure of the phases rather than to the concentration of the dopants such as Ni (at least in the usual commercial range of concentration for duplex stainless steel). In order to verify this, and also to generate EELS spectra for each phase in an independent non-experimental fashion, DFT calculations were carried out.

The spectra obtained for BCC and FCC iron phases by means of DFT calculations are displayed in Fig. 3. When comparing to the spectra in Fig. 1f and Fig. 2d, a striking resemblance can be observed between the calculated spectrum for the BCC phase and the measured one for the α phase. The same occurs for FCC's calculation and the experimental low-loss spectra from γ phase. As is verified from the data in Table 3, the maxima in the EELS spectrum and the respective simulation are in very good agreement. However, the shape of the simulated spectra does not reproduce completely well the one from experiments. Despite this drawback, DFT simulations are still an adequate method to model plasmon behavior due to their high accuracy to predict where the plasmon peak should be located for each phase (disregarding the Ni content of the studied alloy) and, thus, provide a reference to compare to the obtained spectra, especially in the case where more than one phase is contributing to the spectrum or if an unknown phase is present.

3.3. Machine learning based algorithms to cluster low-loss EELS spectra

Fig. 4a shows an HAADF image (rectangular section from Fig. 1a, marked with purple dashed line) and Fig. 4c displays the corresponding EELS spectra acquired along an α/γ interface boundary, both from 2005 as-cast. The plasmon peaks of the α and γ phases are well separated by about 2 eV, the plasmon peak of α phase being situated at lower energy.

Fig. 4b presents a two-region map obtained by k-means clustering, while Fig. 4d displays the corresponding cluster centers. The procedure performed by k-means decides, in first place, which EELS spectra are sufficiently similar to each other and so they are averaged to form a cluster center like those in Fig. 4d. Later on, a map (like the one in Fig. 4b) is drawn comparing the EELS spectra from each (x,y) position to the cluster centers. The algorithm computes point by point a comparison between the experimental EELS spectrum and the cluster centroids. Then, each point is allocated to the cluster having a cluster center closer to the EELS spectrum at that specific point. The algorithm converges when the cluster centroids stabilize and no longer change with each iteration. Our results shows that the clustering can reproduce really well the α and γ domains (corresponding to cluster 0 and 1, respectively) and there is a nearly perfect match between the cluster centers and the representative EELS spectra.

It is worth noting that classical approaches like using a pixel-by-pixel fit leads to a plasmon map which can also be used to discriminate the α and γ domains (Fig. S1, left). However, this approach is much more computational demanding. For comparison, k-means clustering on a

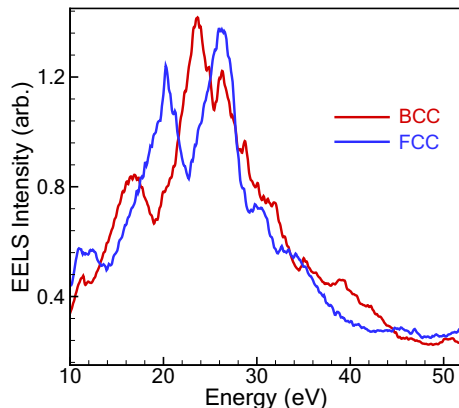


Fig. 3. DFT calculated spectra for $(\text{Im}[-1/\epsilon(\omega)])$ for BCC and FCC iron.

Table 3

Energy position of the maxima (E_p) obtained by each of the considered methods, for both major phases.

Method	$E_p^{\text{Lower E Peak}}$ (eV)	$E_p^{\text{Higher E Peak}}$ (eV)
Low-loss EELS (average)	$E_p^{\alpha} = 24.15 \pm 0.30$	$E_p^{\gamma} = 26.18 \pm 0.20$
DFT	$E_p^{\text{BCC}} = 23.94 \pm 0.05$	$E_p^{\text{FCC}} = 26.25 \pm 0.08$

given dataset and computer was performed in about 30s whereas for the pixel-by-pixel fitting it took about 10 min to complete the task under the same set-up conditions (dataset and computer). In addition, the fitting approach is also time-consuming for the user as it requires a fine pre-tuning of the fitting parameters.

Fig. 5a shows the superposition of Cr elemental map and dark field image (DF) along an α/γ interface boundary of an aged specimen from 2205 alloy. Precipitates along the α/γ interfaces with a size of about 200 nm can be observed. They present a higher Cr concentration and belong to the σ phase which is often observed along the α/γ interfaces after the aging process.

The graphic shown in Fig. 5b is obtained from the silhouette metric, which is a method to determine how adequate is to group the data into a certain number of clusters. In this procedure, each cluster is represented by a “silhouette” and a plot displays how well the entire dataset distributes within each configuration, presenting a maximum for the optimal number of clusters. The silhouette metric plot (Fig. 5b) indicates that three clusters is the optimal choice for this dataset whereas only two were indicated for the as-cast dataset (Fig. S2a).

Fig. 5d and Fig. 5c present the cluster centers and clusters map, respectively. There is an excellent agreement between the results of the clustering and the EDS mapping.

It must be outlined here, that the presence of a third phase having an intermediate energy for volume plasmon peak, as is the case for σ phase precipitates, prevent the pixel-per-pixel method from properly separate the domains (Fig. S3). This confirms that automatic phase separation in DSS can successfully and more efficiently be performed by combining low-loss EELS experiments and k-means clustering than with other techniques.

It should be mentioned that in the case of Fig. 4, where only two phases are present and the EELS spectrum in each phase is unambiguously determined, it is not necessary to perform any intermediate step to correlate each type of cluster to a given phase. Particularly, it is straightforward to associate “cluster center 0” (green curve in Fig. 4d) to low-loss EELS spectrum from α phase (red line in Fig. 4c) and the same happens for “cluster center 1” with the spectrum from γ phase (blue line in Fig. 4c).

However, it is not the same situation when trying to relate low-loss EELS spectra if one or more of the EELS spectra belong to an unknown phase, or if an EELS spectrum might be the result of merged spectra, as occurs in the sample in Fig. 5a. When small precipitates or regions are observed (i.e., sizes below 200 nm) the adjacent phases might be underneath these precipitates and a mixed signal from the precipitate and the neighbouring phase will create a spectrum that is not purely from none of both phases. The only way to provide a proper reference spectrum to compare to “cluster center 2” would be by obtaining one from simulations. It should be noted that the presence of oxide at the surface of the sample as well as interface plasmons could bring more complexity to this kind of study. This is true in particular for really thin samples and/or for higher-resolution studies in which the influence of interface plasmons cannot be discarded. Although, it should be noted that surface and interface plasmons tends to appear at much lower energy in the EELS spectrum than for volume plasmon [33].

4. Conclusions

Ascribing a domain to a certain phase and mapping the phase distribution inside a duplex stainless steel (DSS) can be a challenging task

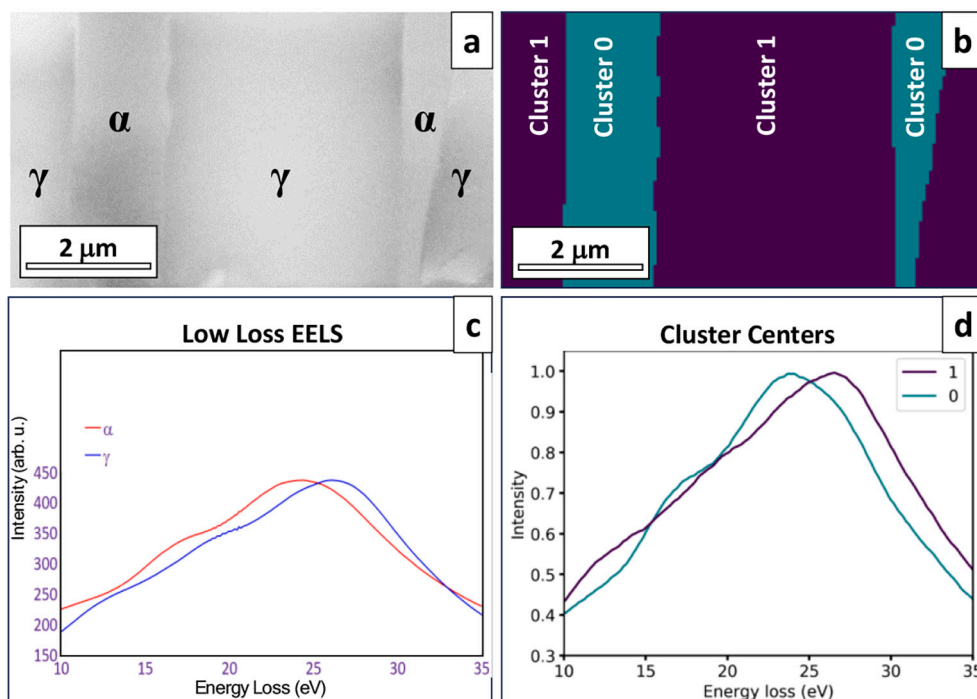


Fig. 4. 2205, as-cast: a) HAADF image acquired simultaneously as the EELS dataset. b) Resulting map obtained by k-means clustering. c) EELS spectra acquired on the α and γ phases. d) corresponding cluster centers.

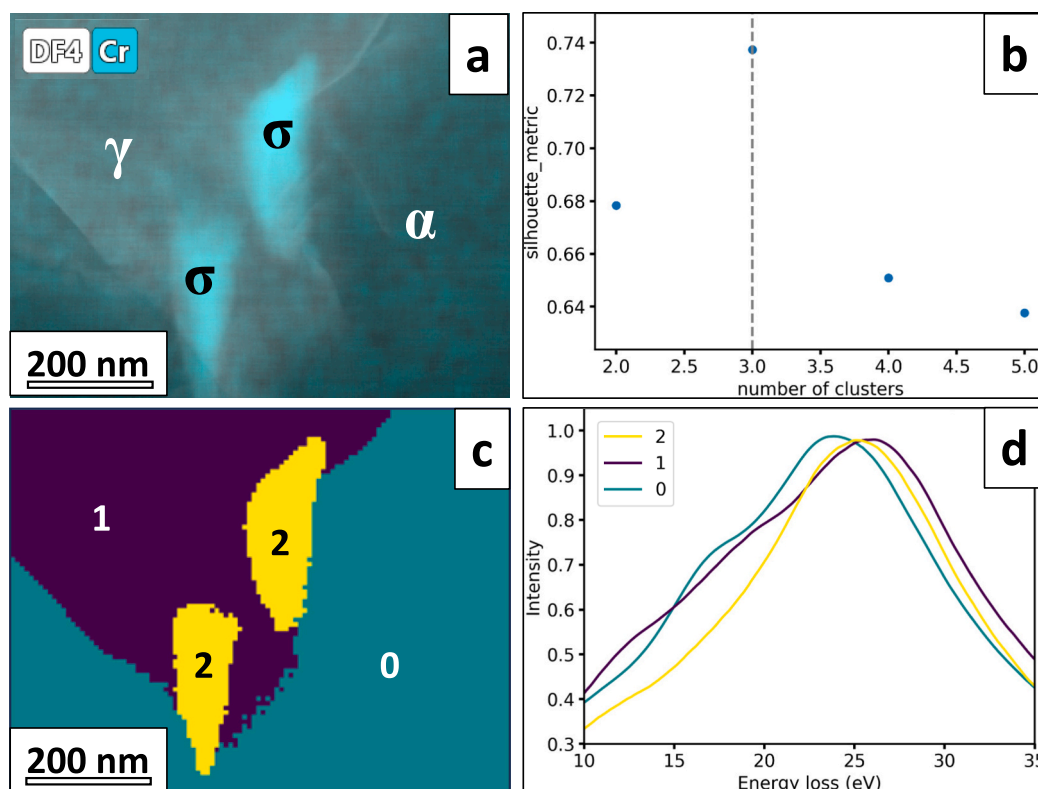


Fig. 5. 2205, aged: a) Superposition of Cr elemental map and DF image. b) Silhouette metric. c) Corresponding clusters maps. d) Cluster centers of k-means clustering.

for which some of the classical approaches fail or are time consuming. Analysing the contrast between grains in HAADF-STEM has proved not to be a trustworthy technique given that the average atomic number (Z) is quite similar for both of the major phases (α -ferrite and γ -austenite).

Thus, for instance, changes in thickness can easily become a more relevant source for intensity variations than Z -contrast, generating changes in contrast that are not related to a phase or compositional transition. Similarity in the values of Z between α and γ is detrimental

for EDS compositional mappings, as well. EDS mappings might be an adequate way to determine which region belong to α or γ but, in order to achieve reliable results, long mapping times are required (around 20 min or more) for small variations of compositions. Studying SAED patterns (or DP) is an effective manner to determine which the phases in presence but the disadvantage is that these SAED must be afterwards compared to simulated DP, which is a tedious and also time-consuming procedure. Additionally, performing the acquisition of SAED patterns in nanometric sized grains (size <100 nm) like those observed for σ phase is not an easy task.

We showed by using DFT that low-loss EELS spectra of DSS are more sensitive to crystallographic changes than to compositional ones and this was an important aspect to evaluate before carrying out the clustering. The energy of the plasmon peak predicted by this method for iron BCC and FCC crystal structures was coincident with the one obtained experimentally for α and γ phases. These results confirmed that the Ni content does not affect the plasmon energy could be used reliably to cluster low-loss EELS dataset in order to differentiate the phases in presence such as α , γ and σ . Applying machine learning clustering algorithms to low-loss EELS spectra allowed us to map the different crystallographic regions accurately and in record time with respect to the other mentioned techniques. Furthermore, clustering method showed to be even more convenient than a classical pixel-per-pixel fitting approach, given that it is capable to distinguish among phases having plasmon energies very close to each other and is much less time-consuming. This technique shows promise for phase studies in stainless steels, including austenitic, ferritic, or duplex varieties. Moreover, its versatility suggests applicability to more intricate systems like corrosion, welding, or deformation studies. Its main advantage is its ease of use and the possibility of coupling it, thanks to the versatility of TEM, with many other analysis techniques such as EELS core-loss analyses, EDS, HRTEM, diffraction... The combination of multiple techniques, whether at the nano, micro or macro level, is still an important point to consider in order to improve the analytical reliability.

CRediT authorship contribution statement

Victoria Castro Riglos: Writing – review & editing, Writing – original draft, Visualization. **Beatriz Amaya Dolores:** Writing – review & editing, Methodology, Investigation, Formal analysis, Data curation. **Ashwin Ramasubramaniam:** Writing – review & editing, Investigation, Formal analysis, Conceptualization. **Lorena González-Souto:** Methodology, Investigation. **Rafael Sanchez:** Funding acquisition. **Javier Botana:** Funding acquisition. **Juan F. Almagro:** Funding acquisition. **José J. Calvino:** Writing – review & editing. **Luc Lajaunie:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The raw data required to reproduce these findings are available from

the corresponding author upon reasonable request.

Acknowledgements

LL acknowledges funding from the Spanish Ministerio de Economía y Competitividad, the Ministerio de Ciencia e Innovación MCIN/AEI/10.13039/501100011033/FEDER UE, and the European Union “NextGenerationEU”/PRTR (PID2022-140370NB-I00, RYC2021-033764-I, CPP2021-008986). The (S)TEM measurements were performed at the National Facility ELECMI ICTS (“Division de Microscopia Electronica”, Universidad de Cadiz, DME-UCA).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.matchar.2024.113924>.

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